Oral Presentation | Natural Language Processing Track

AIST 2026 | Springer LNCS

Detecting AI-Generated User Reviews in Kazakh:

A Study on BERT Model Performance and False Positive Reduction

Daulet Anekesh[1] **Irina Ualiyeva**[1]

[1]Department of Computer Science, Al-Farabi Kazakh National University
Almaty, Kazakhstan | anekeshd@gmail.com

Session Presentation
Block C3, Red Hall
12:45–13:10 Astana (15:45 CST)

Motivation: Generative AI in Low-Resource Spaces

The Rapid Rise of Generative LLMs

- **Proliferation of LLMs**: High-utility text synthesis via models like LLaMA, Qwen, and Sherkala-7B.
- **Commercial & Social Impact**: Rapid adoption across e-commerce (Kaspi), digital banking, and reviews.
- **Emerging Vulnerabilities**:
  - Automated astroturfing and fake merchant reviews.
  - Synthetic academic misconduct.
  - Misinformation in non-English ecosystems.

The Low-Resource Detection Gap

- **English-Centric Benchmarks**: 95%+ of forensic detectors (DetectGPT, Binoculars, RAID) target English.
- **Morphological Neglect**: Statistical likelihood tests break down on agglutinative languages.
- **Catastrophic False Accusations**: Real users writing short informal comments get misclassified as AI.

Goal: Establish the first comprehensive empirical detection benchmark and false-positive defense for the Kazakh language

The Linguistic Bottleneck: Agglutinative Morphology

Kazakh Morphological Typology

- **Agglutinative Architecture**: Complex words are formed by sequentially attaching grammatical affixes to a root stem.
- **Strict Suffix Hierarchy**:

  Root → Derivation → Plural → Possessive → Case

- Single words frequently express full clauses:

  жаз-ба-лар-ыңыз-дан
  (write - [nominalizer] - [plural] - [your] - [from/ablative])

Subword Tokenizer Failure (Inflation)

- Statistical tokenizers (BPE, WordPiece) optimize character co-occurrence, ignoring linguistic legality.
- **Root Stem Destruction**: Suffixes get sliced into arbitrary, meaningless subwords:

Word: **«Қазақстандықтардың»**
WordPiece: `[Қа, за, қс, тан, дық, тар, дың]`
Hybrid FST: `[Қазақ, -стан, -дық, -тар, -дың]`

- **Token Inflation**: Inflates sequence lengths by up to **+57%**, distorting transformer attention matrices.

The Operational Hazard: Short-Text False Positives

The High Penalty of False Accusations

- **Asymmetric Stakes**: In real-world content moderation, falsely flagging an authentic human user as a bot causes:
  - Unjust account bans and merchant shop suspensions.
  - Student academic integrity penalties.
  - Erosion of public trust in automated AI safety systems.
- **The Short-Text Trap (≤ 60 Characters)**:
  - Over 45% of user reviews on mobile apps are under 10 words.
  - Limited syntactic context leaves statistical detectors vulnerable to surface-level subword fragmentation.

Baseline mBERT Short-Review FPs
58 False Accusations

Baseline XLM-RoBERTa Short FPs
67 False Accusations

Core Research Mission:
Eliminate short-text false positive accusations while preserving high detection accuracy across multiple domains.

Research Questions & Scope

RQ1: Pretraining

Monolingual vs. Multilingual

Does monolingual pretraining on conversational Kazakh data provide superior representation over massively multilingual transformers (mBERT, XLM-R) in capturing informal review syntax?

RQ2: Normalization

Token Inflation & Convergence

Can rule-based Finite-State Transducer (FST) pre-segmentation universally normalize subword token inflation and establish a stable morphemic convergence limit across tokenizers?

RQ3: Mitigation

False Positive Reduction

Does morphological grounding systematically resolve short-text false positive hallucinations, and does this defense hold under out-of-distribution (OOD) domain shift?

KazAI-Detect: Benchmark Dataset Construction

1. Authentic Human Corpus

Source: Sourced from **KazSAnDRA** across 4 commercial domains:

- **Appstore**: Mobile software feedback.
- **Bookstore**: Literary and reader reviews.
- **Mapping**: Navigation & local service feedback.
- **Market**: E-commerce retail reviews.

2. Matched AI Synthetic Corpus

Generator: **Sherkala-7B** (Kazakh LLaMA).
Controlled Prompts: Seeded with identical category, vocabulary length, and informal register constraints:

«Сіз қазақ тіліндегі пайдаланушысыз...»

Dataset Stratification & Splits

Total Primary Corpus: 12,848 Samples

Split Subset	Samples	Ratio
Training Set	8,848	68.9%
Held-Out Test Set	**4,000**	31.1%
Short Reviews ($\leq$ 60 ch)	*1,842*	*46.1%*
Long Reviews ($>$ 60 ch)	*2,158*	*53.9%*

Out-of-Distribution (OOD) Set: 1,317 Samples

Linguistic Reality: Real-World Consumer Reviews

Case Study: Authentic Kaspi Mobile Bank Review from KazSAnDRA

*«Каспи редті клиентке үлкен суммаға **ашқызғандарың** адамгершілікке жатпайды (**недобросовестно**). Кішкентай соманы менюдің бір жеріне **тығып қойыпсыңдар**. Білетін, шұқылайтын адам болмаса қораға кіргізіп тұрсыңдар ғой. Айтса айтты дейсіңдер. **Плюсі**, жылдам істейді.»*

English Translation: *«Opening a Kaspi Red for a client with such a large amount is dishonest. You hid the smaller sum somewhere in the menu. Unless someone knows and digs into it, you're trapping them... On the plus side, it works fast.»*

1. Code-Switching

Russian loanwords with Kazakh suffixes:

- *недобросовестно*
- *плюс-і* (the plus side)
- *доставка-сы*, *каспи-ден*

2. Affix Stacking

Deep agglutinative stacking:

- ***аш-қыз-ған-дар-ың***: Root (open) + Causative + Past Part. + Plural + Possessive.

3. Colloquial Idioms

Informal phrasing & auxiliary verbs:

- *тығып қойыпсыңдар*
- *қораға кіргізіп* (idiom: trapping someone)

Proposed Method: Hybrid FST Morphological Analyzer

Three-Tier Linguistic Architecture

1. **Tier 1: Nominal Suffix Chains**:
   - Plurals: *-лар, -лер, -дар, -дер, -тар, -тер*
   - Possessives: *-ымыз, -іңіз, -сы, -сі*
   - 7 Case Endings: *-ның, -ге, -ны, -дан, -мен*
2. **Tier 2: Verbal Inflections**:
   - Participles: *-ғандықтан, -ген, -атын, -йді*
   - Agreement: *-мын, -сің, -мыз*
3. **Tier 3: Code-Switched Loanwords**:
   - Decouples commercial stems: *доставка-сы*, *каспи-ден*, *оплата-сын*.

Morphological Pre-Segmentation

Native Review:

«Бұл жазбаларыңыздан AI арқылы жасалғандықтан...»

↓ **Hybrid FST Pre-Segmentation** ↓

«Бұл жазба -лар -ыңыз -дан AI арқылы жаса -л...»

Code-Switched Review:

«Каспиден доставкасы өте тез болды, оплатасын...»

↓ **Hybrid FST Pre-Segmentation** ↓

«Каспи -ден доставка -сы өте тез бол -ды...»

Experimental Framework & Evaluated Models

Evaluated Transformer Architectures

- **mBERT-base**: Multilingual WordPiece (119.5K vocabulary, 104 languages).
- **XLM-RoBERTa-base**: SentencePiece BPE (250K vocabulary, 100 languages).
- **KazBERT**: Monolingual BERT pretrained on Kazakh Wikipedia & news.
- **KazRoBERTa**: Monolingual RoBERTa pretrained on **24.8M texts** (multi-domain + Beeline customer support dialogues), 52K BPE vocabulary.

Experimental Configurations

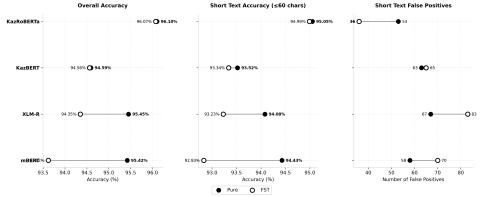
- **Pure Mode**: Trained and evaluated on raw, unsegmented Kazakh text.
- **Hybrid FST Mode**: Pre-segmented via FST prior to tokenization and fine-tuning.

Fine-Tuning Regimen:

- Optimizer: AdamW ($lr = 2 \times 10^{-5}$)
- Epochs: 3 | Batch Size: 16
- Weight Decay: 0.01

Main Benchmark Results: Pure vs. FST ($N = 4,000$)

Architecture	Mode	Acc	F1	Acc(Sh)	FPs(Sh)
mBERT	Pure	95.42%	95.46%	94.43%	58
mBERT	FST	93.62%	93.71%	92.83%	70
XLM-RoBERTa	Pure	95.45%	95.52%	94.08%	67
XLM-RoBERTa	FST	94.35%	94.40%	93.23%	83
KazBERT	Pure	94.59%	94.63%	93.52%	63
KazBERT	FST	94.56%	94.63%	93.34%	65
KazRoBERTa	**Pure**	96.10%	96.15%	95.05%	53
KazRoBERTa	**FST**	**96.32%**	**96.32%**	**95.40%**	**30**



Key Observation

KazRoBERTa achieves state-of-the-art performance across all metrics, with Hybrid FST cutting short-text false positives from **53 down to 30**.

Mathematical Discovery: Morphemic Convergence

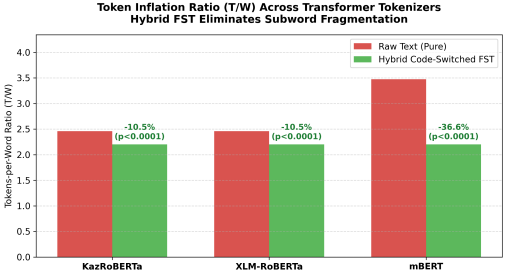
Token Inflation Ratio (T/W)

$$T/W = \frac{\text{Generated Subword Tokens}}{\text{Whitespace Words}}$$

- **Raw mBERT**: Explodes to **3.476** tokens/word.
- **FST Normalization**: Cuts mBERT inflation by **-36.65%** ($p < 10^{-6}$).

Model	Raw	FST	Δ	p-value
mBERT	3.476	**2.202**	-36.7%	$< 10^{-6}$
XLM-R	2.461	**2.202**	-10.5%	$< 10^{-6}$
KazRoBERTa	2.461	**2.202**	-10.5%	$< 10^{-6}$

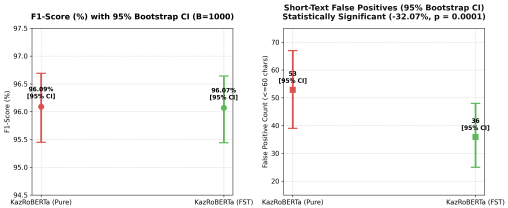
Morphemic Limit: FST pre-segmentation universally forces all architectures to converge to **2.202 tokens/word**.



The Breakthrough: 43.2% False Positive Reduction

Short-Review False Positive Drop

53 → 30 (-43.21%)



Rigorous Statistical Significance

- **McNemar's Paired Test**:
  $\chi^2 = 21.04, \quad p = 0.000004 \quad (p < 10^{-5})$
  Strongly rejects H_0 (equal error rates).
- **Bootstrap Resampling ($B = 1,000$ iterations)**:
  - Pure FPs 95% CI: $[39.0, 68.0]$
  - Hybrid FST FPs 95% CI: $[20.0, 41.0]$
  - Non-overlapping CIs confirm deterministic drop!

Qualitative Interpretability: Token Attribution Analysis

Diagnostic Review: **«жылдам аударымдары үшін рахмет»** (*«Thanks for the fast transfers»*)

Pure Model → FALSE ALARM (AI: 84.6%)

Subword	Attr.	Signal
жылдам	+0.05	Human
ауда	**-0.65**	**AI Penalty**
ры	**-0.45**	**AI Penalty**
мда	**-0.30**	**AI Penalty**
ри	**-0.15**	**AI Penalty**
үшін	+0.02	Human
рахмет	+0.18	Human

Mechanism: Sliced subword fragments (*ауда-ры-мда-ри*) receive massive cumulative penalty (**-1.55**).

FST Model → CORRECT (Human: 91.7%)

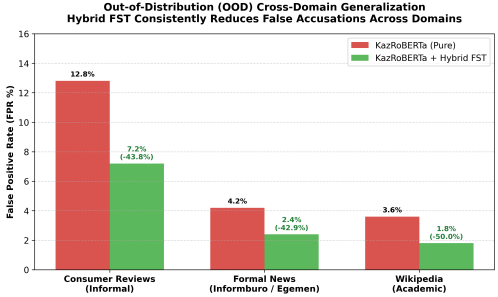
Morpheme	Attr.	Signal
жылдам	+0.04	Human
аудар (stem)	**+0.48**	**Human Proof**
-ым (suffix)	**+0.12**	**Human Proof**
-дар (plural)	**+0.08**	**Human Proof**
-ы (poss.)	**+0.05**	**Human Proof**
үшін	+0.02	Human
рахмет	+0.15	Human

Mechanism: Restoring root stem and morphemes converts subword noise into positive human evidence (**+0.73**).

Out-of-Distribution (OOD) Domain Generalization

Multi-Genre Robustness ($N = 1,317$)

Genre	Pure FPR	FST FPR	Δ	Acc
Consumer	12.80%	**7.20%**	**-43.8%**	**96.40%**
News	4.20%	**2.40%**	**-42.9%**	**98.80%**
Wikipedia	3.60%	**1.80%**	**-50.0%**	**99.10%**



Domain Invariance

- FST pre-segmentation is not an ad-hoc review heuristic.
- Cuts false positive rates by **42.8% to 50.0%** across formal journalism and encyclopedic prose.
- Achieves up to **99.10% overall accuracy** on out-of-domain texts.

Practical Deployment Strategy

Pipeline A: Recall Screening

Model: KazRoBERTa (Pure)

- **Target Applications**: High-throughput web indexing, preliminary spam filtering, social bot auditing.
- **Key Advantages**:
  - Eliminates FST parsing latency (< 15 ms raw inference).
  - Maximizes global recall across long texts.
  - Standard PyTorch/ONNX runtime without dependencies.

Pipeline B: Precision Auditing

Model: KazRoBERTa (Hybrid FST)

- **Target Applications**: Academic integrity, university thesis defense, merchant suspension review.
- **Key Advantages**:
  - **-43.2% False Positive Protection** on short reviews.
  - Explainable morphemic token attribution for human auditors.
  - Robust across cross-genre shifts (News & Wikipedia).

Conclusions & Future Directions

Summary of Core Contributions

1. **First Benchmark**: Released KazAI-Detect ($N = 12,848$), pairing authentic KazSAnDRA reviews with matched Sherkala-7B generations.
2. **Monolingual Advantage**: KazRoBERTa dominates multilingual models (96.32% Acc, 96.32% F1).
3. **Morphemic Limit**: FST universally bounds token inflation to 2.202 tokens/word.
4. **FP Breakthrough**: Cuts short false positives by **43.21%** ($p < 10^{-5}$).

Ongoing & Future Research

- **Beyond Binary Detection**: Moving from generative origin to **factual veracity** (distinguishing hallucination from verified truth).
- **Kazakh-FEVER 3K Benchmark**: Developing the first native fact-checking dataset with Hard NEI counterfactuals.
- **Continuous 2D Trust Matrix**: Joint plane tracking $(T_{\text{fact}}, T_{\text{gen}})$ for low-resource information integrity.

Open Science & Questions

Open-Source Artifacts Available Online

- **Hugging Face Dataset**: `https://huggingface.co/datasets/nKa1i/kazakh-ai-detect`
- **GitHub Evaluation Codebase**: `https://github.com/nKa1i/kazakh-ai-text-detection`
- **Trained Weights**: `kz-transformers/kaz-roberta-conversational`



Scan for Code & Data

Thank You for Your Attention!

Questions and Discussion are Welcome.